

A new geospatial vector data matching algorithm optimised for building change detection

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11/03/2025

Journée de la Recherche UGE-IGN-ENSG 2025

Session: Détection de Changement

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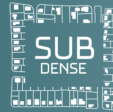
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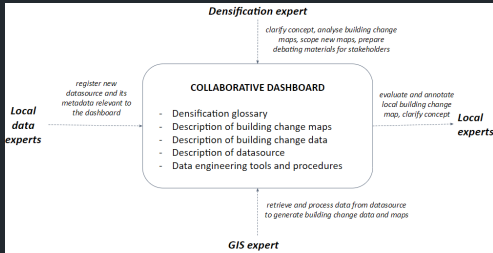
The SubDense European project studies the dynamics of suburban densification by:



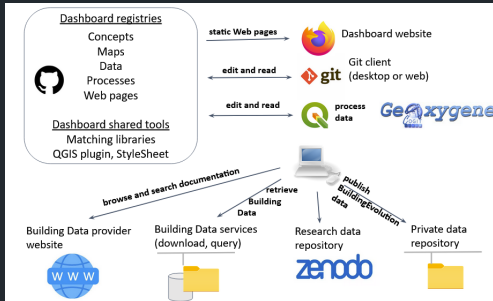
Understanding polyrationalities
of space, actors and policies
on suburban densification

- exploring how diverse strategies of land policy interact with landowners' and local stakeholders' interest and agency to shape suburban densification and their impact on suburbia across different planning systems (France, Germany, UK);
- combining quantitative approaches (geodata analysis and geosimulation) with qualitative approaches (social and policy science and planning).

A collaborative dashboard for mediation



→ git-based dashboard to share knowledge and resources between different types of experts [Bucher et al., 2024]



→ reproducible tools and methods, including **building change detection scripts**

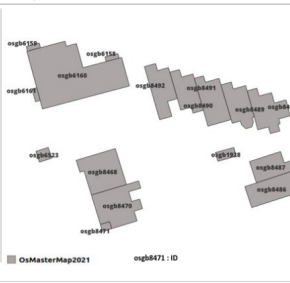
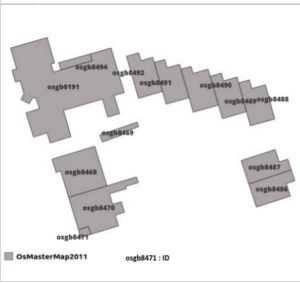
Difficulties with vector data for building change



Orthophoto 2007



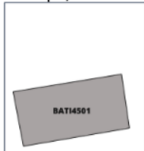
Orthophoto 2015



Example in the UK where changes in OSMasterMap building data includes both real changes and data changes. Source: [Bucher et al., 2025]

Difficulties with vector data for building change

BDTopo, 2011



Orthophoto, 2011

BDTopo, 2021



Orthophoto, 2021

OSMasterMap, 2011



Orthophoto, 2007

OSMasterMap, 2021



Orthophoto, 2015

Example of data specification changes only, for France (BDTopo) and the UK (OSMasterMap). Source: [Bucher et al., 2025]

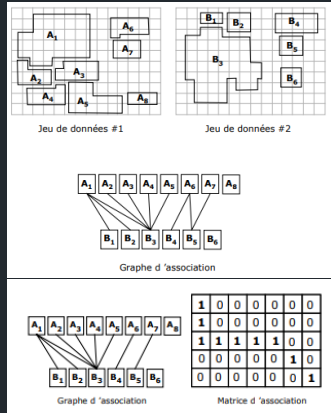
→ Geospatial vector data matching algorithms as a method for change detection [Xavier et al., 2016], applied to linear data [Costes et al., 2015] and polygonal data [Gregory and Healey, 2007].

→ Several algorithms which have been developed and tested for different application (change detection, data quality) and different contexts, with meta-parameters for each.

Research objective:

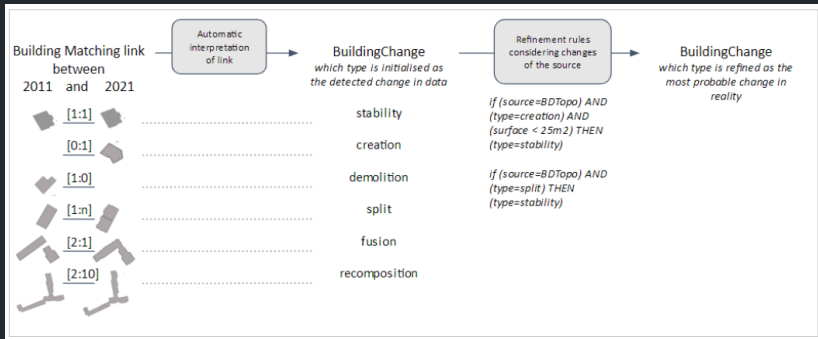
Which algorithm and parametrisation are optimal to detect true building changes, across the three case study countries and the various urban contexts within the study areas?

Algorithm to produce m-n links based only on geometries [Harvey et al., 1998]
[Bel Hadj Ali, 2001]:



1. Construct all possible association links as surfaces with a non-empty intersection (top left Fig.)
2. Filter links with an intersection surface below a threshold parameter
3. Filter links with an intersection surface too small relatively to matched surfaces (rate parameter)
4. Construct all m-n links as connected clusters in the remaining bipartite graph (bottom left Fig.)

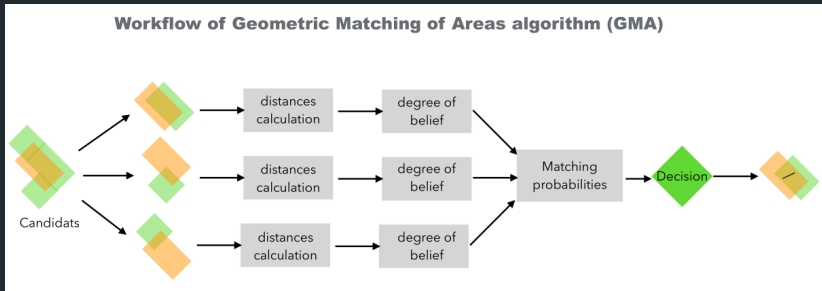
→ implemented in the geoxygene java library [Bucher et al., 2012] by [Mustière, 2002]



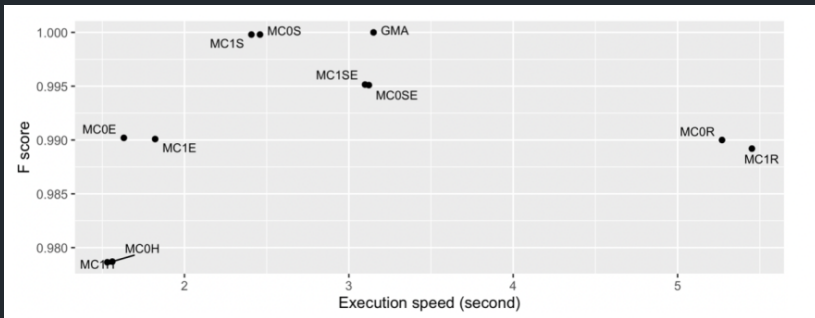
Automatic interpretation of matching links into building change, with specific adjustments for BDTopo [Bucher et al., 2025], by a python script available at <https://github.com/subdense/matching>



Building evolution for two examples in the urban area of Strasbourg
[Bucher et al., 2025]



Implementation of the Multi-criteria matching algorithm [Olteanu-Raimond et al., 2015] in python by [Guardiola et al., 2024a], with euclidian, surface, radial and Hausdorff distances, and run twice to obtain m-n links.



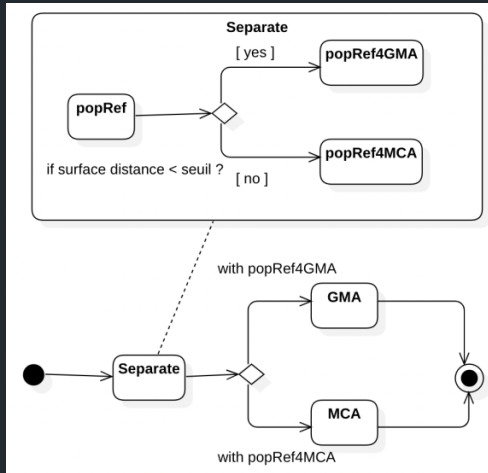
Bi-objective benchmark optimising for performance (F-score on a $\simeq 2k$ buildings ground truth dataset) and runtime, for different parametrisations of the two algorithms, by [Guardiola et al., 2024a]

→ various algorithm performances

→ under-detection of change by GMA and over-detection by MCA

→ GMA better for m-n links, MCA better for 1-1 links

A new algorithm using multi-modeling



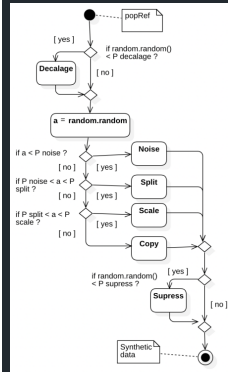
Proposition of a new algorithm combining GMoA and MCA, with the choice made through thresholding surface distance [Guardiola et al., 2024b]



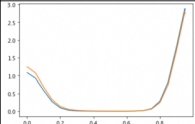
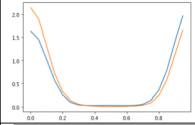
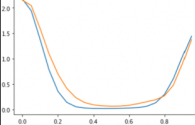
Results obtained on an example in the suburbs of Rennes
[Guardiola et al., 2024b]

Validating the algorithm with synthetic data

	popRef	popCreate
Decalage		
Noise		
Split		
Scale		
Copy		
Suppress		

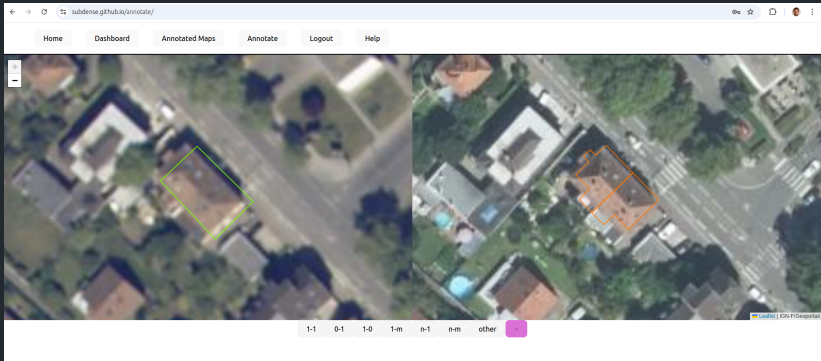


Synthetic data generator introduced by [Guardiola et al., 2024b] to generate datasets on which to validate and optimise the algorithm

Lieu Géographique	Valeur des paramètres optimisés	Courbes des densité des distances surfacique de popRef et popCreee
Centre Urbain	Decallage= 0.05 Brouilleur = 0.45 Echelle = 0.20 Spltieur = 0.25 Copieur = 0.10 Suppression = 0.10	
Banlieue	Decallage= 0.10 Brouilleur = 0.75 Echelle = 0.10 Spltieur = 0.05 Copieur = 0.10 Suppression = 0.10	
Ville périphérique	Decallage= 0.10 Brouilleur = 0.25 Echelle = 0.40 Spltieur = 0.15 Copieur = 0.15 Suppression = 0.10	

Integration into the OpenMOLE software and calibration of the generator for different urban typologies (single fitness: distance between surface distance densities) [Guardiola et al., 2024b]

Validating the algorithm with ground truth data



Online standalone javascript app (deployed automatically via github pages) to annotate potential matching links (what link should be obtained; which real-world change), developed by Julien Perret (<https://github.com/subdense/annotate>)

→ work in progress: large annotation campaign for the 6 case study cities in the 3 countries

Contributions:

- a benchmark of algorithms [Guardiola et al., 2024a]
- a new algorithm combining GMoA and MCA [Guardiola et al., 2024b]
- open source python library <https://github.com/umrlastig/pymatch>
- web app to annotate links <https://github.com/subdense/annotate>

Work in progress:

- annotation campaign to produce a large scale ground-truth dataset
- optimisation using the ground truth in OpenMOLE
- systematic exploration and validation using the synthetic data generator
- include other matching algorithms in the benchmark

Perspectives:

- extension to line and points matching algorithms
- use for data quality issues



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